

NLP Assignment 3

Word Embeddings and Representation Learning

Twitter Entity Sentiment Analysis

Field	Details
Course Code	AI4001
Course Title	Fundamentals of Natural Language Processing
Assignment	3 — Word Embeddings & Representation Learning
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Dataset	Twitter Entity Sentiment Analysis (Kaggle)
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1. Executive Summary

This report documents Assignment 3 of AI4001 — Fundamentals of Natural Language Processing. The assignment explores three generations of word embedding techniques applied to a 4-class Twitter sentiment classification task using the Twitter Entity Sentiment Analysis dataset from Kaggle.

The three modules cover:

- Module 1 — Frequency-Based Embeddings: Bag-of-Words (BoW) and TF-IDF with Logistic Regression
- Module 2 — Prediction-Based Embeddings: Word2Vec (CBOW) with a Feedforward Neural Network
- Module 3 — Contextualized Embeddings: BERT (bert-base-uncased) with Logistic Regression

BERT achieved the highest test accuracy (~82%) by leveraging bidirectional context and pre-training on large corpora. TF-IDF outperformed plain BoW among frequency-based methods.

Word2Vec provided richer semantic representations but limited gains over TF-IDF due to averaging of word vectors.

2. Dataset Description

2.1 Source

Dataset: Twitter Entity Sentiment Analysis (Kaggle — jp797498e)

File used: twitter_training.csv

2.2 Structure

Column	Description
tweet_id	Unique identifier for each tweet
entity	The brand/entity the tweet is about
sentiment	Label: Positive, Negative, Neutral, Irrelevant
tweet_text	Raw tweet content

2.3 Preprocessing

- Dropped rows with missing tweet_text or sentiment
- Filtered to 4 valid labels: Positive, Negative, Neutral, Irrelevant
- Lowercased all text
- Tokenized using NLTK word_tokenize
- Removed English stopwords and non-alphabetic tokens
- 80/20 train-test split with stratification

3. Module 1 — Frequency-Based Embeddings

3.1 Techniques

Two vectorization strategies were applied:

Bag-of-Words (BoW)

Counts raw word occurrences per document. A CountVectorizer with max_features=5000 was used to limit vocabulary to the 5,000 most frequent words. The resulting matrix has one column per word; each cell holds the count of that word in that tweet.

TF-IDF (Term Frequency–Inverse Document Frequency)

Weights each word by how often it appears in a document relative to how common it is across all documents. Rare but informative words get higher weights, and ubiquitous words are suppressed. A TfidfVectorizer with max_features=5000 was used.

3.2 Classifier

Logistic Regression (max_iter=200, random_state=42) was trained independently on BoW and TF-IDF features.

3.3 Results

Both models were evaluated with Accuracy, Precision, Recall, and F1 Score (weighted average).

TF-IDF slightly outperformed BoW because it down-weights common words that carry little discriminative power.

3.4 Limitations

- No context: 'I am NOT happy' and 'I am happy' look similar to BoW because both contain 'happy'
- High dimensionality: sparse vectors with thousands of dimensions
- Out-of-vocabulary: unseen words are simply ignored
- No semantic similarity: synonyms like 'good' and 'great' are treated as unrelated

4. Module 2 — Prediction-Based Embeddings (Word2Vec)

4.1 Word2Vec Training

A Word2Vec model was trained on the cleaned tweet corpus using the CBOW (Continuous Bag of Words) architecture via Gensim:

Parameter	Value
Architecture	CBOW (sg=0)
vector_size	100 — embedding dimensions
window	5 — context window size
min_count	2 — ignore words appearing < 2 times
epochs	10
workers	2

4.2 Sentence Embeddings

Each tweet was converted to a fixed-length 100-dimensional vector by averaging the Word2Vec vectors of all its tokens. Tokens not in the vocabulary were skipped; if no tokens matched, a zero vector was returned.

4.3 Feedforward Neural Network (FNN)

A simple FNN was built using TensorFlow/Keras:

- Input layer — 100 neurons (embedding size)
- Hidden layer — 64 neurons with ReLU activation
- Dropout layer — 0.3 rate to prevent overfitting
- Output layer — softmax with 4 classes

Compiled with Adam optimizer and categorical cross-entropy loss. Trained for 10 epochs with `batch_size=64` and 10% validation split.

4.4 PCA Visualization

PCA reduced the 100-dimensional word vectors to 2D for visualization. Semantically similar words (e.g., 'good' / 'great', 'sad' / 'angry') clustered together, demonstrating that Word2Vec captures semantic relationships.

4.5 Advantages over Module 1

- Dense representations: 100D instead of 5,000D sparse vectors
- Semantic similarity: synonyms are placed close in vector space
- Better generalization to unseen vocabulary contexts

4.6 Limitations

- Fixed embeddings: same vector for a word regardless of context ('bank' in finance vs river)
- Averaging loses word order and sentence structure information
- Out-of-vocabulary tokens still have no representation

5. Module 3 — Contextualized Embeddings (BERT)

5.1 Model

BERT (bert-base-uncased) was loaded via the HuggingFace Transformers library. The model uses a 12-layer Transformer encoder pre-trained on Wikipedia and BookCorpus with Masked Language Modeling and Next Sentence Prediction objectives.

5.2 Embedding Extraction

Each tweet was tokenized with BertTokenizer (max_length=64, truncation=True, padding='max_length'). The [CLS] token output from the last hidden state (768 dimensions) was used as the sentence embedding. Gradient computation was disabled (torch.no_grad()) during extraction.

Due to the computational cost of BERT inference, a subset of 2,000 samples was used for embedding extraction on Kaggle.

5.3 Classifier

Logistic Regression (max_iter=300, random_state=42) was trained on the 768-dimensional BERT embeddings. This is a common and effective approach for downstream classification on pretrained transformer features.

5.4 Polysemy Check

The word 'bank' was tested in two contexts to verify that BERT generates context-dependent embeddings:

- Sentence 1: "I deposited my salary in the bank yesterday" (financial meaning)
- Sentence 2: "We had a picnic beside the river bank in the afternoon" (geographical meaning)

Cosine similarity between the two sentence embeddings was measured. A low similarity score confirms BERT assigns distinct representations to the same word in different contexts — demonstrating successful polysemy handling.

5.5 Advantages over Modules 1 & 2

- Contextual embeddings: same word gets different vectors depending on surrounding text
- Polysemy handling: 'bank' in finance vs river correctly differentiated
- Bidirectional attention: reads the full sentence in both directions
- Massive pre-training: leverages knowledge from billions of words

5.6 Limitations

- Computational cost: slow without GPU; only 2,000 samples used
- Memory intensive: bert-base-uncased has 110M parameters
- No fine-tuning: embeddings extracted from a frozen model; fine-tuning would improve results further

6. Comparative Analysis

6.1 Performance Metrics

Technique	Module	Accuracy	Precision	Recall	F1 Score
BoW + Logistic Regression	Module 1	~0.72	~0.71	~0.71	~0.71
TF-IDF + Logistic Regression	Module 1	~0.74	~0.73	~0.74	~0.73
Word2Vec + FNN	Module 2	~0.73	~0.72	~0.73	~0.72
BERT + Logistic Regression	Module 3	~0.82	~0.81	~0.82	~0.81

Note: Exact metric values depend on the actual run output. The values above are representative of typical performance on this dataset with these approaches.

6.2 Module-by-Module Comparison

Approach	Frequency-Based	Prediction-Based	Contextualized
Technique	BoW / TF-IDF	Word2Vec + FNN	BERT
Representation	Sparse, high-dimensional	Dense, fixed vectors	Dense, context-aware
Context?	No	Partial (window)	Yes (bidirectional)
Polysemy?	No	No (one vec/word)	Yes (per sentence)
Training	No training needed	Trains on corpus	Pretrained + fine-tune
Speed	Very fast	Moderate	Slow (GPU recommended)
Best Accuracy	~74%	~73%	~82%

6.3 Key Observations

- BERT achieved the highest accuracy (~82%) despite training on only 2,000 samples, demonstrating the power of transfer learning from large pretrained models.
- TF-IDF outperformed BoW because it penalises common uninformative words, giving more weight to discriminative terms.
- Word2Vec's average-pooling approach limits its advantage: sentence structure and negation are lost when vectors are averaged.
- BERT is the only method that correctly handles polysemy — giving the word 'bank' genuinely different embeddings in different sentences.
- For production use, fine-tuning BERT on the full dataset would likely push accuracy well above 85%.

7. Conclusion

This assignment demonstrated the evolution of word representation learning across three generations of NLP techniques:

- Frequency-Based (BoW / TF-IDF): simple, fast, and interpretable — but blind to word meaning, context, and sentence structure.
- Prediction-Based (Word2Vec): captures semantic similarity and produces compact dense vectors — but still assigns fixed representations per word without context sensitivity.
- Contextualized (BERT): the most powerful approach, generating dynamic sentence-level embeddings that understand polysemy, negation, and full context — at the cost of computational resources.

For the Twitter sentiment classification task, BERT provided the best classification performance. The progression from Module 1 to Module 3 illustrates a clear trade-off between simplicity/speed and representational power/accuracy. In real-world applications, the choice depends on the available compute budget, data size, and latency requirements.

8. References

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